STA 013 — Elementary Statistics

Problem Set — Day 6

Variance & Standard Deviation · Shortcut Formula · Tchebysheff's Theorem & Empirical Rule · Z-Scores · Quartiles, IQR & Box Plots

Based on Lecture: Day 6 · Professor Max · Summer Session I

Name: ______________________ Student ID: ______________________

Section: ______________________ Date: ______________________

Instructions: *This problem set covers Day 6 material only: variance and standard deviation (definition and shortcut formulas), Tchebysheff's Theorem, the Empirical Rule, z-scores, and quartile/IQR/box-plot construction. For written-response and multi-part problems, show all work in the space provided—a correct final answer with no supporting work will receive partial credit only. For multiple-choice items with a "Justify" line, the justification is required for full credit. Unsimplified fractions are acceptable unless otherwise noted. Calculators are permitted.*

Part I — Variance and Standard Deviation

1. A sample data set has n = 5 observations: 5, 12, 6, 8, 14 (the flower-petal data from lecture). [16 pts]

(a) Compute the sample mean **x■**.

Work:

(b) Fill in the deviation table below, then compute the sample variance s^2 using the **definition formula**: $s^2 = \Sigma(x_i - x\blacksquare)^2 / (n-1)$.

i	x■	x■ – x■	(x■ – x■)²
1	5		
2	12		
3	6		
4	8		
5	14		

Σ	—	—	

Compute s² from the table:

(c) Now use the **shortcut formula** to verify your answer: $s^2 = [\ \Sigma x_i^2 - (\Sigma x_i)^2/n\ ] / (n-1)$

Work:

2. You are given: $\Sigma x_i^2 = 10$, $\Sigma x_i = 2$, $n = 10$. Use the shortcut formula to find s^2 and s. [8 pts]

Work:

3. Multiple Choice. Which of the following correctly states why the shortcut variance formula and the definition formula always give the same numerical answer? [3 pts]

(A) They are different approximations that happen to agree when n is large.

(B) They are algebraically identical — the shortcut is obtained by expanding $(x_i - \mu)^2$ and simplifying, so they compute the exact same quantity.

(C) The shortcut formula only works when the sample mean equals zero.

(D) The definition formula is for populations; the shortcut is for samples.

Answer: ______ **Justify in one sentence:**

Part II — Tchebysheff's Theorem and the Empirical Rule

4. A data set has mean $\mu = 45$ and standard deviation $\sigma = 8$. Use Tchebysheff's Theorem to answer the following. [10 pts]

(a) What is the minimum proportion of measurements that must fall within the interval [29, 61]? Show how you identify k, then apply the formula.

Work:

(b) A classmate says "at least 89% of the data is within 3 standard deviations of the mean." Is this correct? Compute the exact Tchebysheff bound for $k = 3$ and confirm or correct the claim.

Work:

(c) Tchebysheff's Theorem gives a *minimum* guarantee. Could the actual proportion within [29, 61] be higher than your answer in (a)? Answer yes or no, and explain why in one sentence.

Answer:

5. A standardized exam score is approximately mound-shaped with $\mu = 100$ and $\sigma = 15$. [12 pts]

(a) What approximate percentage of scores falls between 70 and 130? Identify which Empirical Rule interval this corresponds to.

Work:

(b) What approximate percentage of scores is **above 115**? Use the symmetry of the bell curve — show your reasoning step by step.

Work:

(c) Multiple Choice. A classmate uses Tchebysheff's Theorem (k = 2) to estimate the proportion within [70, 130] and gets "at least 75%." Your Empirical Rule answer from (a) says approximately 95%. Which is correct?

(A) Both are wrong — you need exact data to compute this.

(B) Both are correct: Tchebysheff gives the minimum guarantee; the Empirical Rule gives a tighter estimate that applies specifically because the distribution is mound-shaped.

(C) Only the Tchebysheff answer is valid; the Empirical Rule cannot be applied here.

(D) Only the Empirical Rule answer is valid; Tchebysheff's Theorem doesn't apply to normal data.

Answer: ______ **Justify in one sentence:**

Part III — Approximating s and Z-Scores

6. The salaries (in thousands of dollars) of 30 employees at a small firm range from a minimum of $22k to a maximum of $70k. [8 pts]

(a) Approximate the standard deviation using **$s \approx R/4$**. Show the computation.

Work:

(b) Now approximate using **$s \approx R/6$**. Which approximation would the professor say is more appropriate here, and why?

Work / Reasoning:

7. Using the box-plot data from lecture: 0, 1, 1, 2, 3, 3, 6, 7, 10, 12 (n = 10), the sample mean is x■ = 4.5 and the sample standard deviation is $s \approx 4.09$. [10 pts]

(a) Compute the z-score for the observation x = 10. Interpret what this z-score means in one sentence.

Work + Interpretation:

(b) Compute the z-score for x = 0. Based on its sign and magnitude, what does this tell you about where 0 sits in the data?

Work + Interpretation:

(c) Multiple Choice. A measurement has a z-score of −2.3. Which statement is correct?

(A) The measurement is 2.3 units above the mean.

(B) The measurement is 2.3 standard deviations below the mean.

(C) The measurement is negative.

(D) The measurement equals −2.3.

Answer: ______

Part IV — Quartiles, IQR, and Box Plots

8. Consider the data set: 3, 7, 8, 10, 12, 15, 42 [20 pts]

(a) The data is already sorted. Using the midpoint rule, find Q1, Q2 (median), and Q3. Show all position calculations.

Work:

(b) Compute the IQR, the Lower Fence, and the Upper Fence.

Work:

(c) Identify any outlier(s). State the value(s) and confirm they fall outside the appropriate fence.

Work:

(d) Draw the box plot on the number line below. Mark Q1, Q2, Q3, the whisker endpoints, and any outliers with *. Remember: whiskers extend to the most extreme data point *inside* the fence, not to the fence itself.

(e) Based on the box plot you drew, is this distribution symmetric, right-skewed, or left-skewed? Justify using the relative lengths of the whiskers and the position of the median within the box.

Answer:

9. Multiple Choice. A box plot is constructed from a data set where Q1 = 20, Q3 = 40, and IQR = 20. The upper fence is computed as 70. The largest data value is 65. Where does the right whisker end? **[4 pts]**

(A) At 70, the upper fence

(B) At 65, the largest data value, since it is inside the upper fence

(C) At 40, since the whisker can never extend past Q3

(D) At 50, the midpoint between Q3 and the fence

Answer: ______ Justify in one sentence:

10. Multiple Choice. A box plot shows that the right whisker is substantially longer than the left whisker, and the median line sits closer to Q1 than to Q3. Which conclusion is best

supported? [4 pts]

(A) The distribution is left-skewed (skewed left).

(B) The distribution is symmetric.

(C) The distribution is right-skewed (skewed right), with a longer tail of larger values.

(D) No conclusion about skewness can be drawn from a box plot.

Answer: ______ **Justify:**

Part V — Synthesis

11. Short Answer. A student computes the variance of a data set and gets $s^2 = 0$. What does this tell you about the data set? Is this possible? Explain in two to three sentences using the definition of variance. [6 pts]

Answer:

12. Short Answer. Explain in three to four sentences why Tchebysheff's Theorem and the Empirical Rule give different percentages for the same interval, and when you should use each one. Reference the shape assumption that determines which tool is appropriate. [6 pts]

Answer:

Answer Key & Full Solutions

Problem Set — Day 6 Material

1.

(a) x■ = (5+12+6+8+14)/5 = 45/5 = **9**.

(b) Deviation table:

i	x■	x■ – 9	(x■ – 9)²
1	5	–4	16
2	12	+3	9
3	6	–3	9
4	8	–1	1
5	14	+5	25
Σ	45	0 (✓)	60

s^2 = 60 / (5–1) = 60/4 = **15**.   s = √15 ≈ **3.873**.

(c) Shortcut: Σx■² = 25+144+36+64+196 = 465. Σx■ = 45. s^2 = [465 – (45)²/5] / 4 = [465 – 405] / 4 = 60/4 = **15** ✓ (matches definition formula).

2.

s^2 = [10 – (2)²/10] / (10–1) = [10 – 0.4] / 9 = 9.6/9 ≈ **1.067**.

s = √1.067 ≈ **1.033**.

3.

Answer: (B). The shortcut is derived by algebraically expanding (x■ – μ)² = x■² – 2μx■ + μ² and summing — after simplification the two expressions are identical, not approximations of each other.

4.

(a) [29, 61] = [μ–16, μ+16]. k = 16/σ = 16/8 = **2**. Tchebysheff: at least 1 – 1/k² = 1 – 1/4 = **3/4 = 75%**.

(b) k=3: 1 – 1/9 = 8/9 ≈ 88.9%. The classmate said "at least 89%" — this is a slight overstatement; the exact Tchebysheff bound is 8/9 ≈ 88.9%, not 89%. The claim is approximately correct but technically imprecise.

(c) Yes. Tchebysheff gives a minimum guarantee only. If the distribution happens to be mound-shaped, the actual proportion within 2σ would be approximately 95% — substantially higher than the 75% minimum.

5.

(a) $70 = \mu - 2\sigma = 100-30$; $130 = \mu+2\sigma = 100+30$. Empirical Rule: $\mu \pm 2\sigma \to \approx$**95%** of scores.

(b) $115 = \mu+\sigma$. The interval $\mu\pm\sigma$ contains ~68%, so 32% lies outside; by symmetry 16% is above $\mu+\sigma = 115$. ≈**16%** of scores are above 115.

(c) Answer: (B). Both tools are correct for their domains. Tchebysheff applies to any distribution and gives a minimum (at least 75%); the Empirical Rule applies specifically to mound-shaped distributions and gives the tighter, more precise estimate of ~95%.

6.

(a) $R = 70 - 22 = 48$. $s \approx R/4 = 48/4 =$ **12**.

(b) $s \approx R/6 = 48/6 =$ **8**. With a range of 48 spanning salary data that likely has considerable spread, R/4 is the more standard starting approximation unless told the range is “large.” The professor will specify which to use — practice both.

7.

(a) $z = (10 - 4.5) / 4.09 \approx$ **1.35**. The value 10 is approximately 1.35 standard deviations *above* the mean.

(b) $z = (0 - 4.5) / 4.09 \approx$ **–1.10**. The negative sign confirms 0 is *below* the mean, and a magnitude of ~1.10 means it is slightly more than one standard deviation below average.

(c) Answer: (B). A z-score measures standard deviations from the mean, not raw units. A $z = -2.3$ means the measurement is 2.3 standard deviations below the mean; the actual value could be positive, negative, or zero depending on the data.

8.

(a) Sorted: 3, 7, 8, 10, 12, 15, 42. $n = 7$.

Q1 pos = 0.25(8) = 2 → position 2 exactly → **Q1 = 7**.

Q2 pos = 0.50(8) = 4 → position 4 exactly → **Q2 = 10**.

Q3 pos = 0.75(8) = 6 → position 6 exactly → **Q3 = 15**.

(b) IQR = 15 – 7 = **8**. Lower Fence = 7 – 1.5(8) = 7 – 12 = **–5**. Upper Fence = 15 + 1.5(8) = 15 + 12 = **27**.

(c) 42 > 27 (upper fence) → **42 is an outlier**. No values fall below the lower fence of –5.

(d) Left whisker: from Q1=7 back to Min=3 (inside fence). Right whisker: from Q3=15 to the last non-outlier value = 15 (Q3 itself is the rightmost non-outlier; the whisker has zero visible length on the right before the gap). Outlier 42 plotted as * beyond the gap.

(e) The distribution is **right-skewed**. The outlier on the right (42) and the gap between the right whisker endpoint and the outlier indicate a long tail of larger values. The median (10) sitting roughly centered in the box is consistent with the skew being primarily driven by the extreme outlier rather than the box itself.

9.

Answer: (B). The whisker extends to the most extreme data point that still falls *inside* the fence (i.e., ≤ upper fence of 70), which is 65. The whisker endpoint is a data point, never the fence line itself.

10.

Answer: (C). A longer right whisker means data trails further to the right, and the median sitting closer to Q1 confirms the bulk of the data is on the lower end — both are hallmarks of right-skew.

11.

Yes, this is possible. $s^2 = 0$ means Σ(x■ – x■)² = 0. Since every squared term is non-negative, the only way their sum is zero is if every term is individually zero — i.e., every x■ = x■. This means all values in the data set are identical (no variability whatsoever).

12.

Tchebysheff's Theorem makes no assumption about distribution shape — it applies to any data set and provides a guaranteed minimum percentage (e.g., at least 75% within 2σ). The Empirical Rule requires the distribution to be approximately mound-shaped (bell-curve); when that condition holds, it gives a much tighter estimate (approximately 95% within 2σ). Use Tchebysheff when shape is unknown or non-normal; use the Empirical Rule when the distribution is explicitly described as mound-shaped or normal.

Study tip: For box-plot problems, always compute Q1, Q2, Q3 and IQR before the fences — an error in any quartile propagates to everything downstream. For Empirical Rule vs. Tchebysheff questions, the first thing to check is whether the problem states the distribution is mound-shaped.